MODULE: METABOLISM

Thermodynamics as applied to biological systems:

The study of energy transfer and transformations that occur in matter is called thermodynamics. There are two laws of thermodynamics which are equally applicable for both animate and inanimate matter.

According to first law of thermodynamics, the total amount of energy in the universe is constant. Energy can neither be created nor destroyed. For example, green plants convert light energy to chemical energy that is stored in the food molecules which they manufacture. In muscle cells, chemical energy is transformed into kinetic energy. In living forms, one kind of energy is transformed into another form of energy.

According to second law of thermodynamics, for any process to occur, there is a decrease in free energy and increase in randomness or disorder in the system. This disorderliness is called entropy. There is always a regular and continuous dissipation of energy from the living system in the form of heat and conversion of heat is non-convertible in living system. Thus, it increases randomness or entropy in the living system.

Endergonic versus exergonic reactions:

Endergonic reactions	Exergonic reactions
They are unfavourable or non-spontaneous reactions.	They are favourable or spontaneous reactions.
They absorb energy from the surroundings.	They release energy to the surroundings.
Free energy of the system increases that is the standard Gibb's free energy (G) is positive.	Free energy of the system decreases that is the standard Gibb's free energy (G) is negative.
Entropy (S) decreases.	Entropy (S) increases.

Exothermic versus endothermic reactions:

Exothermic reaction	Endothermic reaction
Enthalpy (heat content) of the product is less than that of the reactants.	Enthalpy (heat content) of the product is greater than that of the reactants.
Products are more stable than reactants.	Reactants are more stable than products.
Heat is evolved during the reaction.	Heat is absorbed during the reaction.
Enthalpy change (ΔH) is negative.	Enthalpy change (ΔH) is positive.
The reaction should be preferably conducted at low temperature for favourable yield of the product.	The reaction should be preferably conducted at high temperature for favourable yield of the product.

Concept of Keq and its relation to standard free energy:

The equilibrium constant of a chemical reaction is the value of its reaction quotient at chemical equilibrium, a state approached by a dynamic chemical system after sufficient time has elapsed at which its composition has no measurable tendency towards further change. For a given set of reaction conditions, the equilibrium constant is independent of the initial analytical concentrations of the reactant and product species in the mixture.

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The equilibrium constant is represented by Keq in an equation. If Keq is greater than 1, it means the concentrations of the products are greater than those of the reactants, and forward reaction is favoured. If Keq is less than 1, it means the concentrations of the reactants are greater than the products, and the reverse reaction is favoured. If Keq is 1, it means neither forward nor is the reverse reaction favoured.
Knowledge of equilibrium constants is essential for the understanding of many chemical systems, as well as biochemical processes such as oxygen transport by haemoglobin in the blood and acid-base homeostasis in the human body.

The free energy change of the reaction in any state, ΔG (when equilibrium has not been attained) is related to the standard free energy change of the reaction, ΔG^0 (which is equal to the difference in free energies of formation of the products and reactants both in their standard states) according to the equation: $\Delta G = \Delta G^0 + RT \ln Q$; where Q is the reaction quotient.
When equilibrium is attained, there is no further free energy change, i.e. $\Delta G = 0$ and Q becomes equal to equilibrium constant (Keq). Hence the above equation becomes:

$$\Delta G^0 = -RT \ln Keq$$
$$\text{or, } \Delta G^0 = -2.303\ RT \log Keq$$

Spontaneity:

The tendency of a process to occur naturally is called its spontaneity. A process accompanied by an increase in entropy tends to be spontaneous. Entropy is a thermodynamic state quantity that is a measure of the randomness or disorder of the molecules of the system. The symbol of entropy is S, while the change in disorder accompanying a process is represented by ΔS. entropy of a system is state function and depends only on initial and final states of the system. Therefore,

$$\Delta S = S_{final} - S_{initial}$$
when, $S_{final} > S_{initial}$ the ΔS is positive.

ATP as an energy currency:

ATP (Adenosine triphosphate) consists of a purine base adenine, a pentose sugar ribose and three molecules of phosphoric acids. The adenine and ribose collectively constitute nucleoside adenosine, which by having one, two or three phosphate groups form AMP (Adenosine monophosphate), ADP (Adenosine diphosphate) and ATP (Adenosine triphosphate) respectively.
In ATP, the last phosphate group is linked with ADP by a special bond known as 'Energy rich bond', because when the last phosphate group of the ATP is released, a large amount of energy is released as shown by the following reaction:

$$A–P\sim P\sim P \rightarrow A–P\sim P + Pi + 7300 \text{ calorie}$$

The utility of energy rich phosphate bond of the ATP is that, great amount of energy is kept stored in the ready state in a very limited space of the cell. This stored chemical energy is disposed very quickly at the time of the need in various cellular functions. Thus ATP is called the 'Energy currency of the cell'.

Breakdown of glucose to $CO_2 + H_2O$ (Glycolysis and Krebs cycle):

1. **Glycolysis:**

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Glycolysis

- It is the process of partial oxidation of glucose or similar hexose sugar into two molecules of pyruvic acid through a series of ten enzyme mediated reactions, releasing energy as ATP and reducing power as $NADH_2$.
- It occurs in cytosol or cytoplasm.
- It is common to both aerobic and anaerobic respiration.
- It is regulated by three enzymes, catalysing non-equilibrium reactions: **hexokinase, phosphofructokinase** and **pyruvate kinase**.

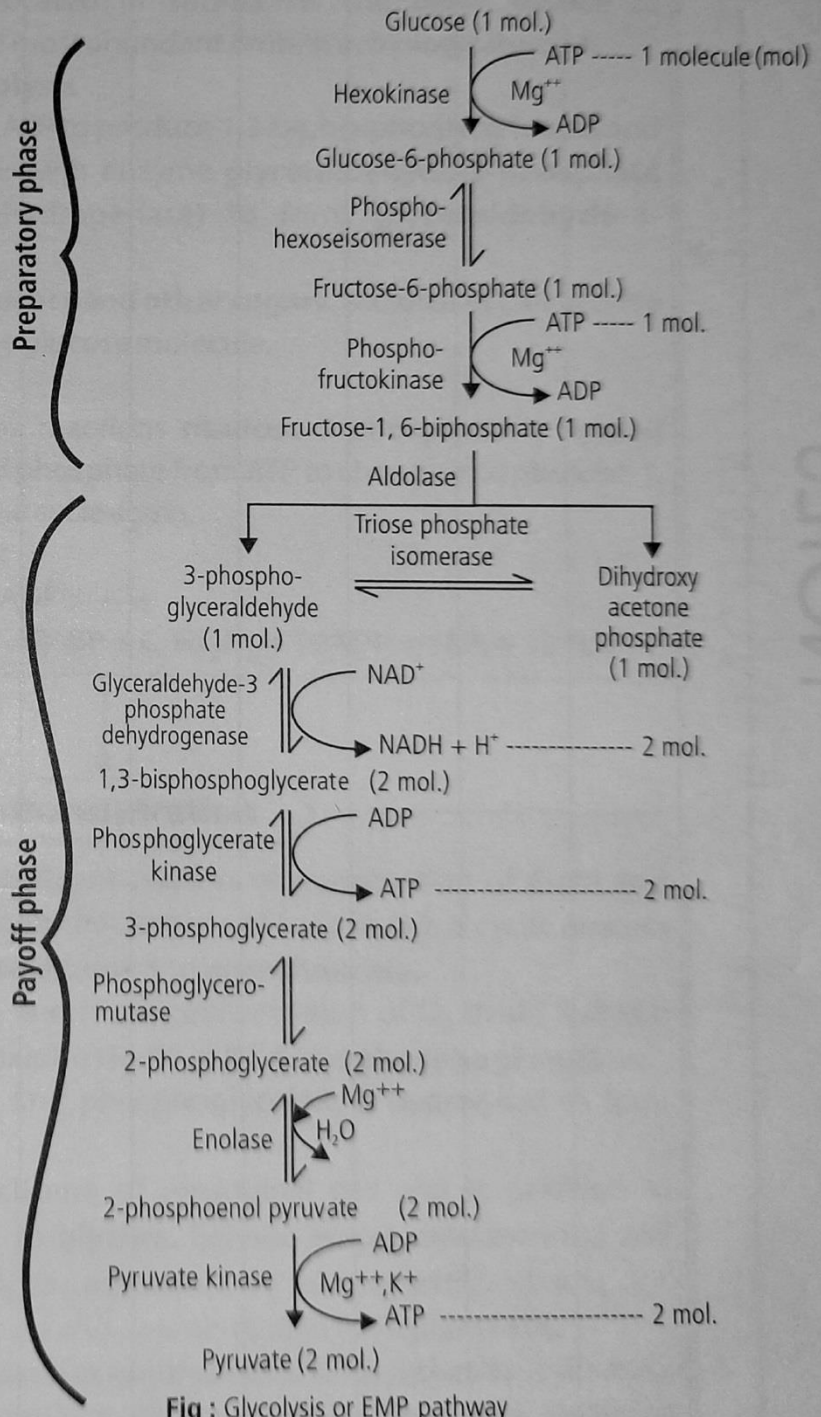


Fig : Glycolysis or EMP pathway

- Cells handle pyruvic acid in 2 different ways in presence of O_2 (aerobic respiration) and in absence of O_2 (anaerobic respiration / fermentation).

2. **Krebs' Cycle (Citric acid Cycle):**

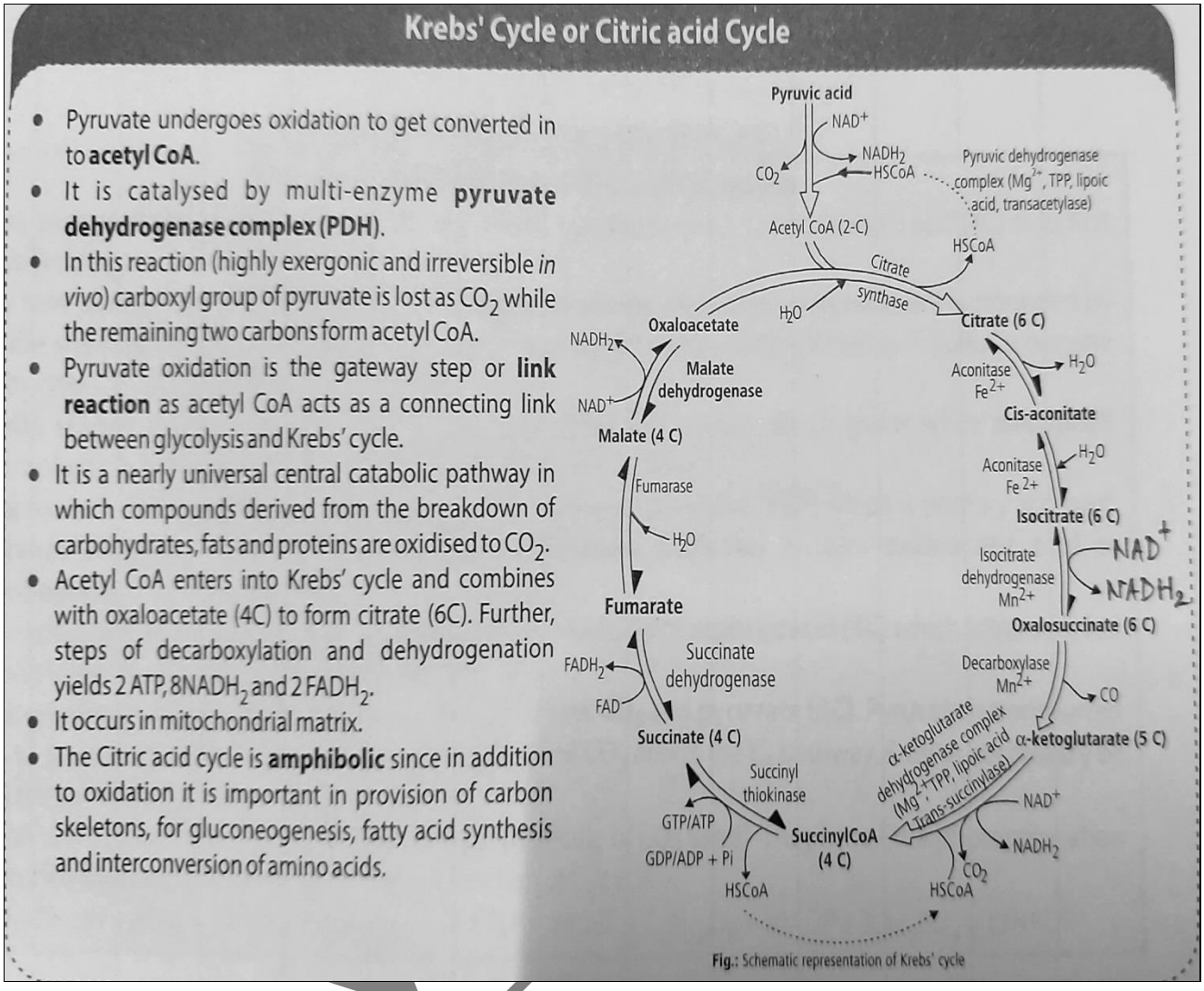
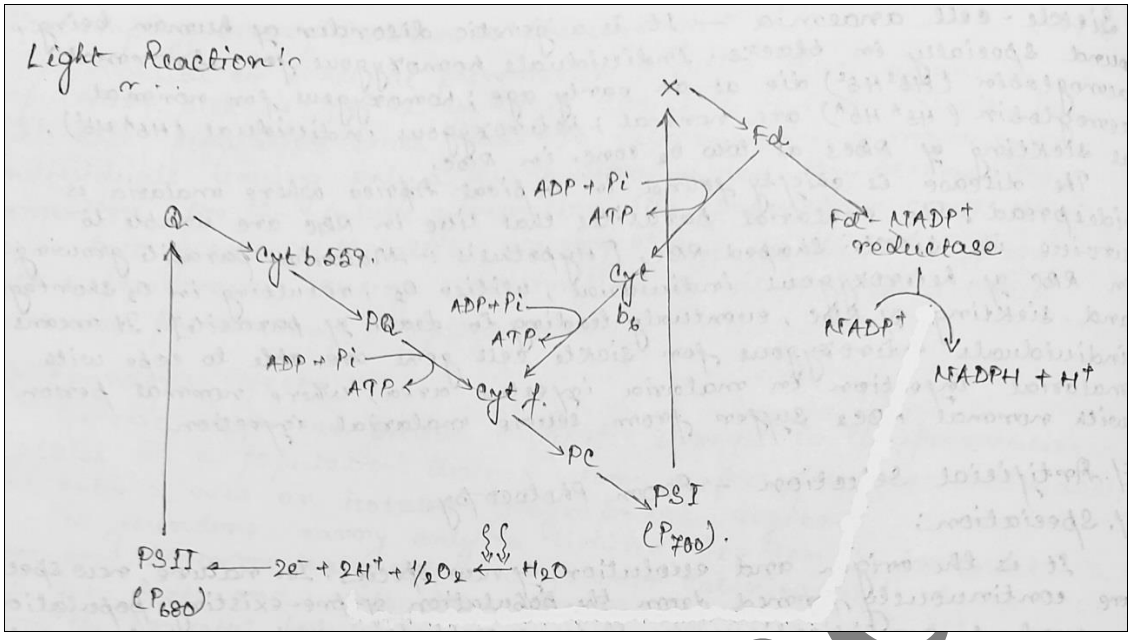


Fig.: Schematic representation of Krebs' cycle

Synthesis of glucose from CO_2 and H_2O (Photosynthesis):

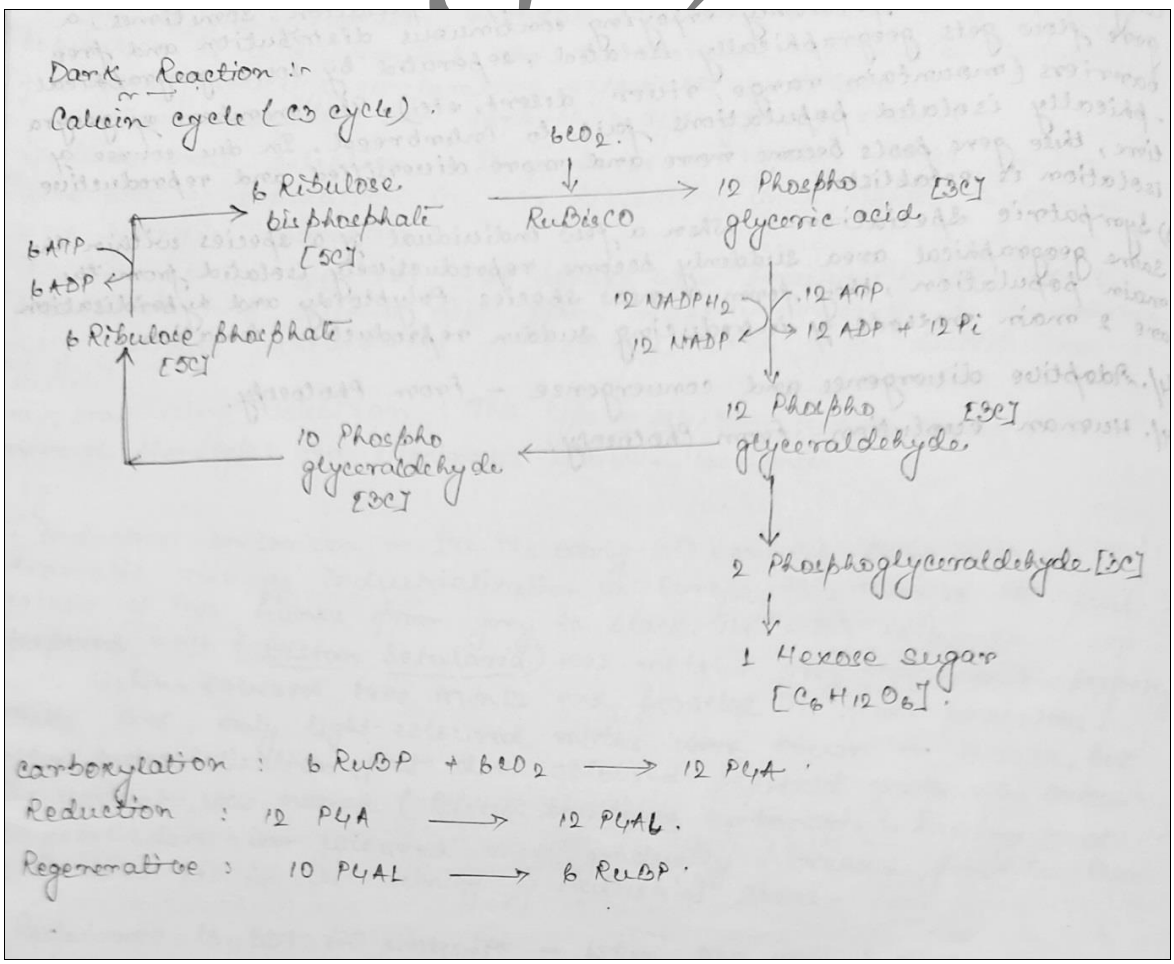
Photosynthesis is an enzyme regulated process for production of organic compound (Glucose) inside the chlorophyll containing cells from CO_2 and H_2O with the help of sunlight. Photosynthesis occurs in two phases, namely, photochemical phase and biosynthetic phase.

1. **Photochemical phase:**
   This is the light dependent phase of photosynthesis occurring inside thylakoids to produce assimilatory power (NADPH and ATP molecules).



2. Biosynthetic phase:

This phase of photosynthesis catalyse assimilation of CO_2 to carbohydrates. The reactions do not require light but assimilatory power produced during photochemical phase. It occurs in the stroma of chloroplast.



Energy yielding and energy consuming reactions:

The work of biological growth depends on the transfer of energy from catabolic (energy yielding) to anabolic (energy consuming) reactions. Anabolic pathways, also called biosynthetic pathways, as expected consume energy. For example, assembly of a protein from amino acids require energy input. On the other hand, catabolic pathways, also called degradative pathways, lead to release of energy. For example, when glucose is degraded to lactic acid in our skeletal muscle, energy is liberated. Living organisms have learnt to trap this energy liberated during degradation and store it in the form of chemical bonds. As and when needed, this bond energy is utilised for biosynthetic, osmotic and mechanical work that we perform.

Concept of Energy charge:

The energy charge is an index used to measure the energy status of biological cells. ATP is the principal molecule for storing and transferring energy in the cell; it is used for biosynthetic pathways, maintenance of transmembrane gradients, movement, cell division etc. More than 90% of the ATP is produced by phosphorylation of ADP by the ATP synthase. ATP can also be produced by substrate level phosphorylation reactions. Adenylate kinase is the enzyme that maintains the three adenine nucleotides in equilibrium.

The energy charge is related to ATP, ADP and AMP concentrations. It was first defined by Atkinson and Walton who found that it was necessary to take into account the concentration of all three nucleotides, rather than just ATP and ADP, to account for the energy status in metabolism.

The energy charge of most cells varies between 0.7 and 0.95 – oscillations in this range are quite frequent. Daniel Atkinson showed that when the energy charge increases from 0.6 to 1.0, the anabolic (ATP-demanding) pathways are activated while catabolic (ATP-supplying) pathways are inhibited. He concluded that control of these pathways has evolved to maintain the energy charge within rather narrow limits. We now know that most (if not all) anabolic and catabolic pathways are indeed controlled directly and indirectly by energy charge.

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